

	Disjunction	Quantification (names)	Quantification (free variables)	Quantification (lambda abstracts)
Inferential role premises	$\vee I$: $[\phi] \vdash [\phi \vee \psi] \quad [\psi] \vdash [\phi \vee \psi]$. $\vee E$: Whenever $[\phi] \vdash [\chi]$ and $[\psi] \vdash [\chi]$, $[\phi \vee \psi] \vdash [\chi]$.	$\exists I$: $[\phi(c)] \vdash [\exists v \phi(v)]$ $\exists E$: When $[\phi(c)] \vdash [\chi]$, and c occurs nowhere in χ , $[\exists v \phi(v)] \vdash [\chi]$.	$\exists I'$: $[\phi(v)] \vdash [\exists v \phi(v)]$ $\exists E'$: When $[\phi(v)] \vdash [\chi]$ and v is not free in χ , $[\exists v \phi(v)] \vdash [\chi]$.	$\exists I''$: $[\lambda v \phi(v)] \vdash [\lambda v \exists v \phi(v)]$ $\exists E''$: When $[\lambda v \phi(v)] \vdash [\lambda v \chi]$, and χ is closed, $[\exists v \phi(v)] \vdash [\chi]$.
Translation premises	(i) $[\phi]_1 \Leftrightarrow [\phi']_2$ (ii) $[\psi]_1 \Leftrightarrow [\psi']_2$ (iii) $[\chi]_1 \Leftrightarrow [\phi' \vee \psi']_2$ (iv) $[\chi']_2 \Leftrightarrow [\phi \vee \psi]_1$	For some c, d : (i) $[\phi(c)]_1 \Leftrightarrow [\phi'(d)]_2$, (ii) $[\exists x \phi(x)]_2 \Leftrightarrow [\chi]_1$ (iii) $[\exists x \phi(x)]_1 \Leftrightarrow [\chi']_2$, where c does not appear in χ and d does not appear in χ' .	For some v, u : (i) $[\phi(v)]_1 \Leftrightarrow [\phi'(u)]_2$ (ii) $[\exists u \phi'(u)]_2 \Leftrightarrow [\chi]_1$ (iii) $[\exists v \phi(v)]_1 \Leftrightarrow [\chi']_2$, where v is not free in χ and u is not free in χ' .	(i) $[\lambda x \phi(x)]_1 \Leftrightarrow [\lambda x \phi'(x)]_2$ (ii) $[\exists x \phi'(x)]_2 \Leftrightarrow [\chi]_1$ (iii) $[\exists x \phi(x)]_1 \Leftrightarrow [\chi']_2$, where χ and χ' are closed.
Argument	(1) $[\phi']_1 \vdash [\phi \vee \psi]_2$ and $[\psi']_1 \vdash [\phi \vee \psi]_2$. (2) $[\phi]_1 \vdash [\chi]_1$ and $[\psi]_1 \vdash [\chi]_1$ (3) $[\phi' \vee \psi']_1 \vdash [\chi]_1$. (4) $[\phi' \vee \psi']_1 \vdash [\phi \vee \psi]_2$. and analogously, $[\phi \vee \psi]_2 \vdash [\phi' \vee \psi']_1$	(1) $[\phi(c)]_1 \vdash [\exists x \phi'(x)]_2$ (2) $[\phi(c)]_1 \vdash [\chi]_1$ (3) $[\exists x \phi(x)]_1 \vdash [\chi]_1$ (4) $[\exists x \phi(x)]_1 \vdash [\exists x \phi'(x)]_2$ and analogously, $[\exists x \phi'(x)]_2 \vdash [\exists x \phi(x)]_1$	(1) $[\phi(v)]_1 \vdash [\exists u \phi'(u)]_2$ (2) $[\phi(v)]_1 \vdash [\chi]_1$ (3) $[\exists v \phi(v)]_1 \vdash [\chi]_1$ (4) $[\exists v \phi(v)]_1 \vdash [\exists u \phi'(u)]_2$ and analogously, $[\exists u \phi'(u)]_2 \vdash [\exists v \phi(v)]_1$	(1) $[\lambda x \phi(x)]_1 \vdash [\lambda x \exists x \phi'(x)]_2$ (2) $[\lambda x \phi(x)]_1 \vdash [\lambda x \chi]_1$ (3) $[\exists x \phi(x)]_1 \vdash [\chi]_1$ (4) $[\exists x \phi(x)]_1 \vdash [\exists x \phi'(x)]_2$ and analogously, $[\exists x \phi'(x)]_2 \vdash [\exists x \phi(x)]_1$
Problems	(A) How do we get from use to conclusions about entailment? (B) Can we get a more general result that applies even when we don't assume intensional equivalence? (C) Does the conclusion really show that the meanings of $\vee$ in L_1 and L_2 are equivalent?	(A)-(C), plus: (D) Problems for $\exists E$: i) $[\text{John is from Germany}] \vdash [\text{John is German}]$ but $[\text{John is from something}] \not\vdash [\text{John is German}]$ ii) $[\text{Mike is a hammer}] \vdash [\text{Head is attached to Handle}]$ but $[\text{Something is a hammer}] \not\vdash [\text{Head is attached to Handle}]$. (E) Problems for $\exists I$: i) Empty names: $[\text{Vulcan is nonexistent}] \not\vdash [\text{Something is nonexistent}]$. ii) Names for contingent existents: $[\text{John is nonexistent}] \not\vdash [\text{Something is nonexistent}]$ (F) What if subject-predicate sentences in one language are not equivalent to subject-predicate sentences in the other?	(A-C) and (E.ii), plus (G) What sort of thing is $[\phi]$ when ϕ contains free variables? How is $[\phi(x)]$ different from $[\phi(y)]$? And given that there must be a difference in these cases, how could the meaning of an open sentence in one language ever be the same as the meaning of an open sentence in a different language?	(A-C) and (E.ii), plus (H) Even if we can get from use to conclusions about entailment among <i>sentences</i> , how do we get from use to conclusions about entailment among <i>predicates</i> ? •If 'F entails G' means something like 'Necessarily everything that is F is G', then variation in the meanings of quantifiers to generate variation in the meaning of 'entails' applied to predicates. •But perhaps entailment among predicates (property identity) can be understood in a way that doesn't bring in quantifiers?